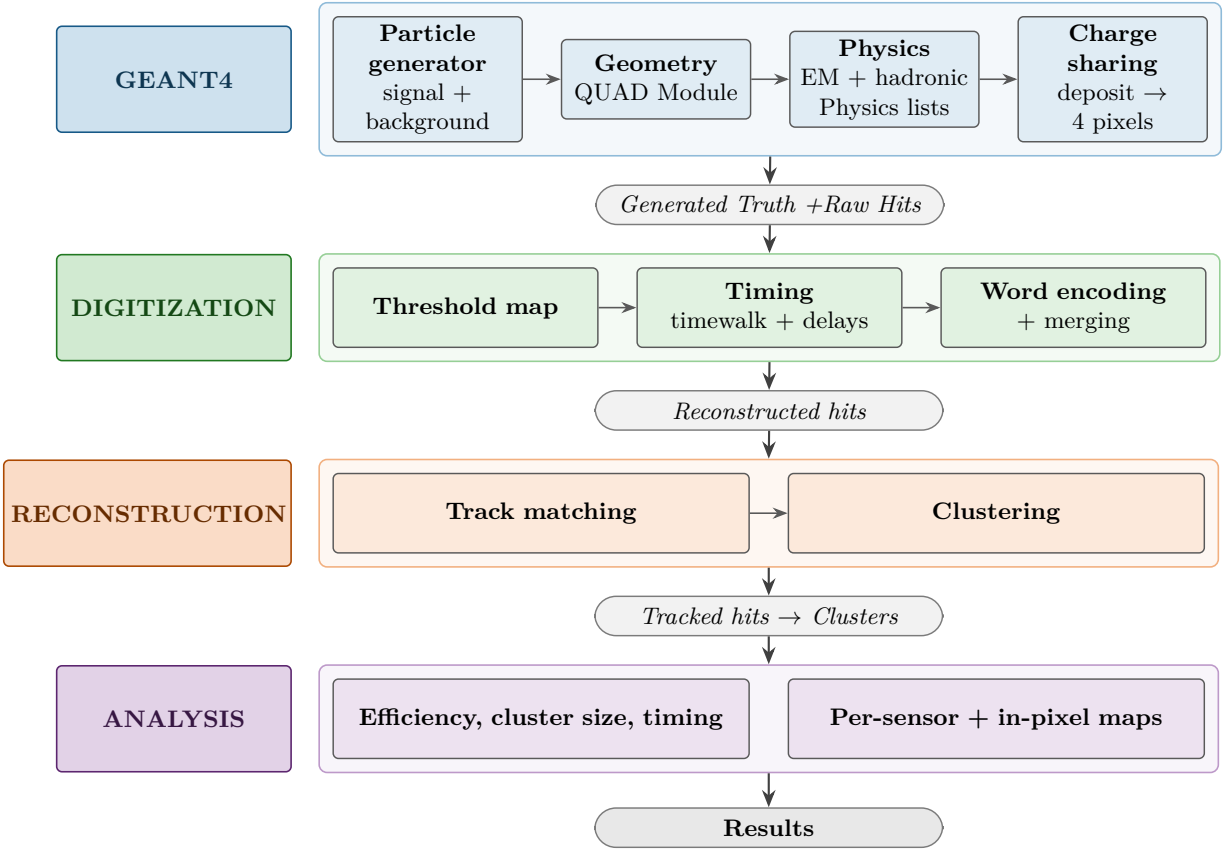

GEANT4
Particle generator
signal + background
Geometry
QUAD Module
Physics
EM + hadronic
Physics lists
Charge sharing
deposit → 4 pixels
Generated Truth +Raw Hits
DIGITIZATION
Threshold map
Timing
timewalk + delays
Word encoding
+ merging
Reconstructed hits
RECONSTRUCTION
Track matching
Clustering
Tracked hits → Clusters
ANALYSIS
Efficiency, cluster size, timing
Per-sensor + in-pixel maps
Results